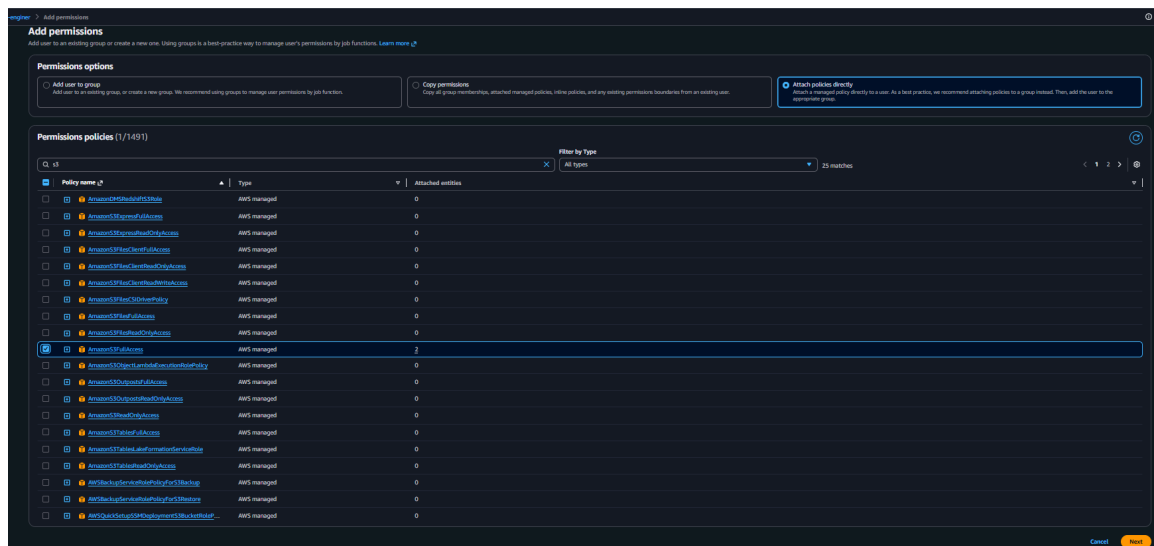
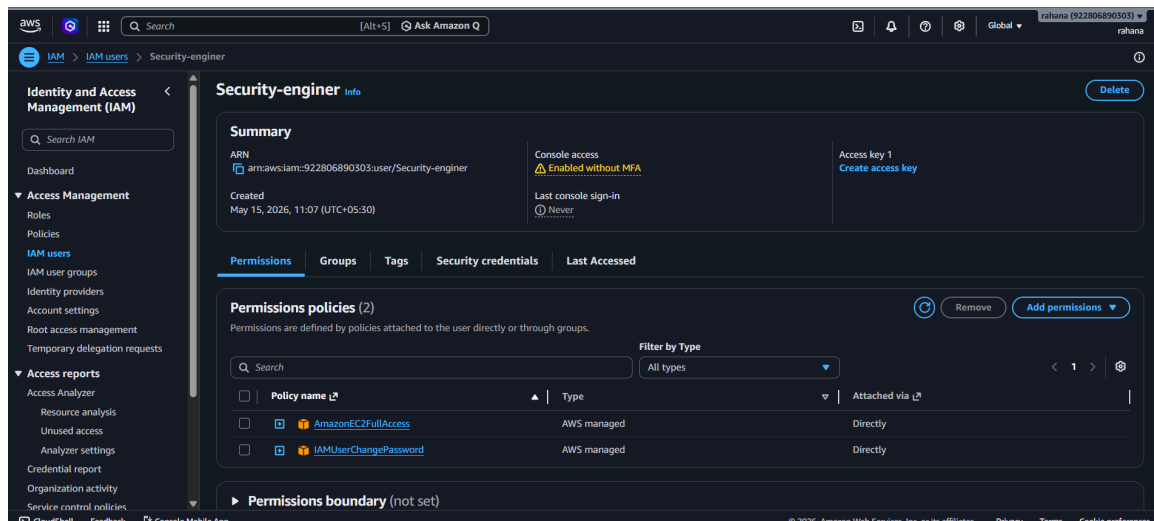


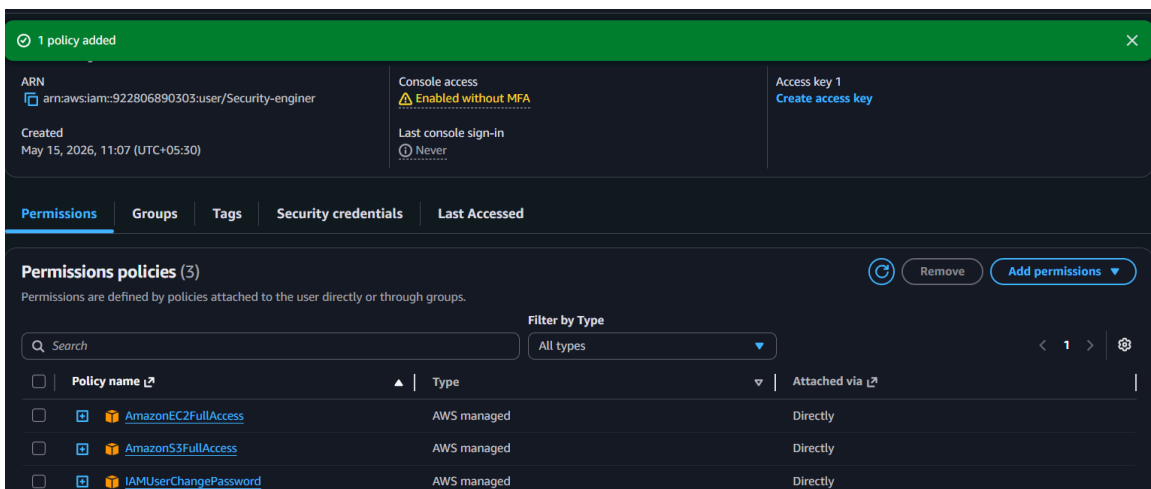
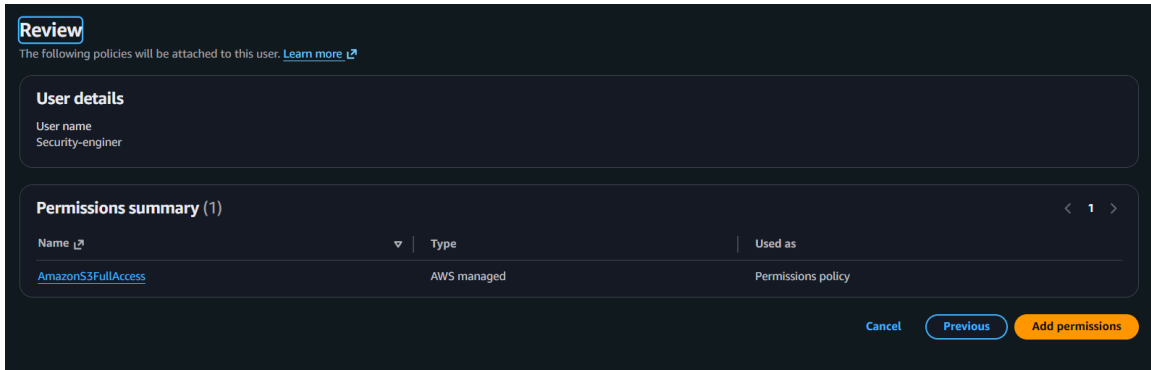
AWS Privilege Escalation Detection Workflow

This report demonstrates the implementation of AWS Privilege Escalation Detection using AWS CloudTrail, CloudWatch Metric Filters, CloudWatch Alarms, SNS Notifications, and Splunk SIEM integration.

Step 1 – Generate Privilege Escalation Event

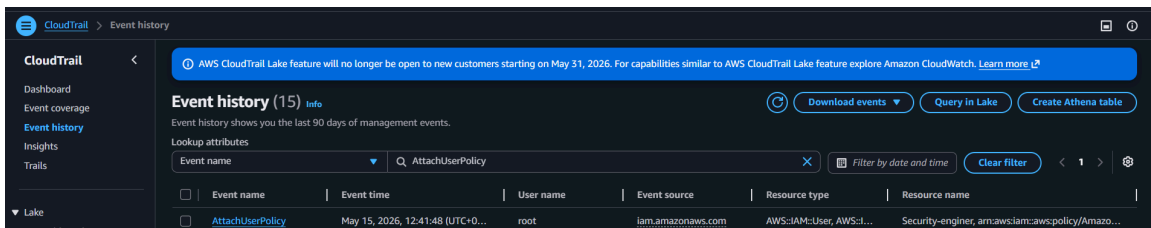
Before creating monitoring and alerting, generate an AttachUserPolicy event by attaching a policy to an IAM user





Step 2 – Verify Event in CloudTrail

Verify that the AttachUserPolicy event is successfully generated inside AWS CloudTrail Event History.



Step 3 – Create CloudWatch Metric Filter

Create a CloudWatch Metric Filter using the AttachUserPolicy event to detect privilege escalation activities.

Log management > aws-cloudtrail-logs-nextwork

< Log streams Tags Data protection Anomaly detection **Metric filters** Subscription filters Contributor Insights Field in >

Metric filters (4)

Find metric filters

1

IAMUserCreationFilter

Filter pattern
{ (\$.eventName = "CreateUser") }

Field selection criteria
-

Metric
[SecurityMonitoring](#) / [IAMUserCreated](#)

Metric value
1

Default value
-

SecretAccessFilter

Filter pattern
{ \$.eventName = "GetSecretValue" }

Field selection criteria
-

Metric
[SecurityMonitoring](#) / [SecretAccessCount](#)

Metric value
1

Default value
-

Step 1
Define pattern
Step 2
Assign metric
Step 3
Review and create

Define pattern

Create filter pattern

You can use metric filters to monitor events in a log group as they are sent to CloudWatch Logs. You can monitor and count specific terms or extract values from log events and associate the results with a metric. [Learn more about pattern syntax.](#)

Filter pattern
Specify the terms or pattern to match in your log events to create metrics.

{ (\$.eventName = "AttachUserPolicy") }

☐ Enable metric filter on transformed logs
When enabled, metric filter will be applied to transformed logs. When disabled, metric filter will be applied to original logs.

Field selection criteria - optional
Filtering options based on account and region provided for centralized log groups

[Search and select Field selection criteria](#)

Builder **Editor**

Test pattern

The test does not include the field selection criteria

Select log data to test
All log data fetched will be in original log format

Custom log data

Test pattern

 The test does not include the field selection criteria

Select log data to test

All log data fetched will be in original log format

Custom log data

Log event messages

Type log data to test with your Filter Pattern. Please use line breaks to separate log events.

```
[B3078518-fcc1-4d30-9573-8b9737671438] BENCHMARK : Running Start Crawl for Crawler
TestCrawler2
[B3078518-fcc1-4d30-9573-8b9737671438] BENCHMARK : Classification complete, writing results to
database mygluedatabase
[B3078518-fcc1-4d30-9573-8b9737671438] INFO : Crawler configured with SchemaChangePolicy
{"UpdateBehavior":"UPDATE_IN_DATABASE","DeleteBehavior":"DEPRECATE_IN_DATABASE"}.
```

Test pattern

Results

Please select log event messages above and click "Test pattern" to see results.

Cancel

Next

Assign metric

Create filter name

Filter name

IAMPrivilegeEscalationFilter

Filter patterns

Filter pattern

Metric details

Metric namespace

SecurityMonitoring

☐ Create new

Metric name

IAMPrivilegeEscalation

Metric value

Metric value is the value published to the metric name when a Filter Pattern match occurs.

Default value – optional

Dimensions

A dimension is a name/value pair that is a part of the identity of a metric. You can assign up to 3 dimensions to a metric. Dimension names and values can be up to 255 characters, cannot contain blank spaces or non-ASCII characters. [Learn more](#)

System fields

Select system fields to include as dimensions

☐ @aws.account

☐ @aws.region

Custom fields

Add up to 3 custom dimension fields

No custom fields added yet

[Add new field](#)

[Cancel](#)

[Previous](#)

[Next](#)

Review and create

Step 1: Pattern

[Edit](#)

Create filter pattern

Filter pattern

```
{ ($.eventName = "AttachUserPolicy") }
```

Step 2: Metric

[Edit](#)

Assign metric

Filter name
IAMPrivilegeEscalationFilter

Metric namespace
SecurityMonitoring

Metric value
1

Unit
-

Metric name
IAMPrivilegeEscalation

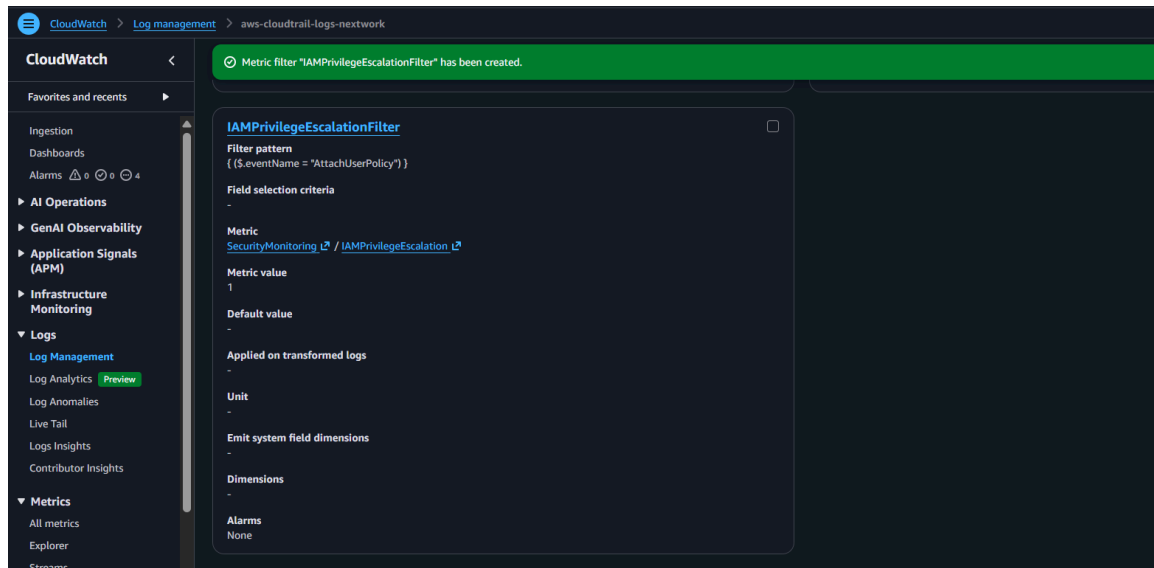
Applied on transformed logs
-

Default value
-

[Cancel](#)

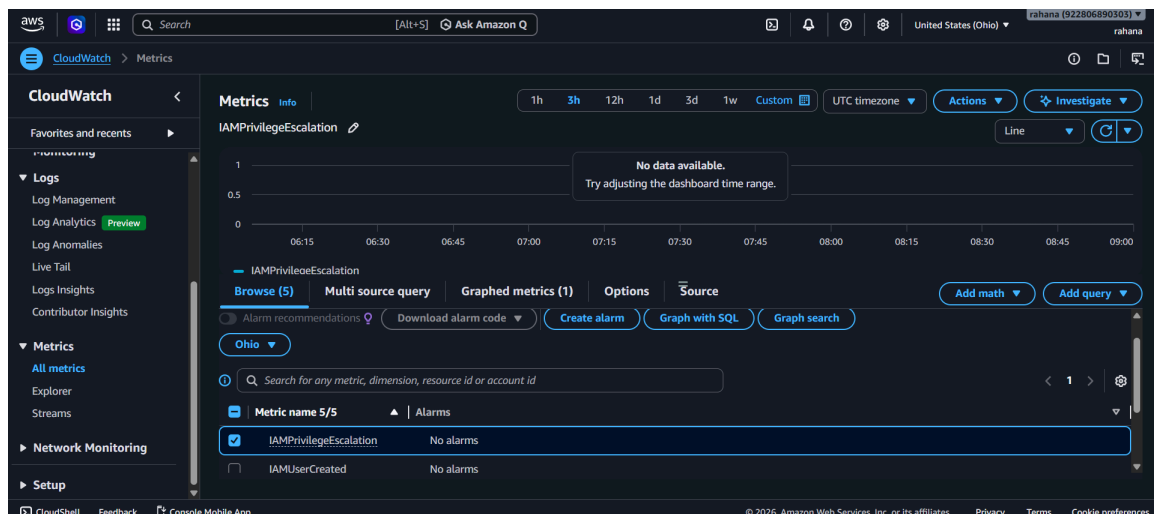
[Previous](#)

[Create metric filter](#)



Step 4 – Validate Metric Generation

Generate another AttachUserPolicy event to verify that the metric filter is functioning correctly and metrics are visible in CloudWatch.



Step 5 – Create CloudWatch Alarm

Create a CloudWatch Alarm for IAMPrivilegeEscalation metric using Sum statistic and threshold value greater than or equal to 1.

Step 1 **Specify metric and conditions**
 Step 2 Configure actions
 Step 3 Add alarm details
 Step 4 Preview and create

Specify metric and conditions

Metric

Graph
This alarm will trigger when the blue line goes above the red line for 1 datapoints within 5 minutes.

1 IAMPrivilegeEscalation

Namespace
SecurityMonitoring

Metric name
IAMPrivilegeEscalation

Statistic
Sum

Period
5 minutes

Conditions

Threshold type

☒ **Static**
Use a value as a threshold

☐ **Anomaly detection**
Use a band as a threshold

Whenever IAMPrivilegeEscalation is...
Define the alarm condition.

☐ **Greater**
> threshold

☒ **Greater/Equal**
>= threshold

☐ **Lower/Equal**
<= threshold

☐ **Lower**
< threshold

than...
Define the threshold value.

1

Must be a number.

► **Additional configuration**

Cancel Next

Step 6 – Configure SNS Notification

Configure Amazon SNS to send real-time email alerts whenever the privilege escalation alarm enters the ALARM state.

CloudWatch > Alarms > Create alarm

Step 1 Specify metric and conditions
 Step 2 **Configure actions**
 Step 3 Add alarm details
 Step 4 Preview and create

Configure actions

Notification

Alarm state trigger
Define the alarm state that will trigger this action.

☒ **In alarm**
The metric or expression is outside of the defined threshold.

☐ **OK**
The metric or expression is within the defined threshold.

☐ **Insufficient data**
The alarm has just started or not enough data is available.

Send a notification to the following SNS topic
Define the SNS (Simple Notification Service) topic that will receive the notification.

☒ **Select an existing SNS topic:**

☐ Create new topic

☐ Use topic ARN to notify other accounts

Send a notification to...

SecurityAlertsbynetwork

Only topics belonging to this account are listed here. All persons and applications subscribed to the selected topic will receive notifications.

Email (endpoints)
nadi.lan23@lpu.in - View in SNS Console

Add notification

CloudWatch

Alarms

Create alarm

Step 1

Specify metric and conditions

Step 2

Configure actions

Step 3

Add alarm details

Step 4

Preview and create

Add alarm details

Name and description

Alarm name

IAM-Privilege-Escalation-Alert

Alarm description - optional

View formatting guidelines

Edit

Preview

This is an H1

double asterisks will produce strong character

This is [an example](https://example.com/) inline link.

Up to 1024 characters (0/1024).

Markdown formatting is only applied when viewing your alarm in the console. The description will remain in plain text in the alarm notifications.

Tags - optional

info

No tags associated with the resource.

Add new tag

You can add up to 50 tags.

Cancel

Previous

Next

Step 1

Specify metric and conditions

Step 2

Configure actions

Step 3

Add alarm details

Step 4

Preview and create

Preview and create

Step 1: Specify metric and conditions

Metric

Graph

This alarm will trigger when the blue line goes above the red line for 1 datapoints within 5 minutes.

1

06:30

07:00

07:30

08:00

08:30

09:00

IAMPriVilegeEscalation

Namespace

SecurityMonitoring

Metric name

IAMPriVilegeEscalation

Statistic

Sum

Period

5 minutes

Conditions

Threshold type

Static

Whenever IAMPriVilegeEscalation is

Greater/Equal (≥)

Step 2: Configure actions

Actions

Notification

When in alarm, send a notification to "SecurityAlertsbynextwork"

Step 3: Add alarm details

Alarm details

Name

IAM-Privilege-Escalation-Alert

Description

-

Tags (0)

Cancel

Previous

Create alarm

Successfully created alarm IAM-Privilege-Escalation-Alert. [View alarm](#) X

Alarms Mute rules - new

Alarms (5)

Filter alarms Quick filters

☐	Name	State	Actions	Actions muted - new	Last state update (UTC)	Conditions
☐	IAM-Privilege-Escalation-Alert	Insufficient data	Actions enabled	-	2026-05-15 09:10:01	IAMPrivilegeEscalation >= 1 for 1 datapoints within 5 minutes
☐	IAM-User-Creation-Alert	Insufficient data	Actions enabled	-	2026-05-15 06:34:53	IAMUserCreated >= 1 for 1 datapoints within 5 minutes
☐	PutSecretValue	Insufficient data	Actions enabled	-	2026-05-12 10:28:53	SecretUpdateCount >= 1 for 1 datapoints within 5 minutes
☐	SecretAccessAlarm	Insufficient data	Actions enabled	-	2026-05-12 10:28:49	SecretAccessCount >= 1 for 1 datapoints within 5 minutes
☐	SecretDeleteAlert	Insufficient data	Actions enabled	-	2026-05-12 09:55:18	SecretDeleteCount >= 1 for 1 datapoints within 1 minute

ALARM: "IAM-Privilege-Escalation-Alert" in US East (Ohio) [Summarize this email](#)

AN AWS Notifications <no-reply@sns.amazonaws.com> Fri 5/15/2026 2:40 PM

To: Nadil AN

You are receiving this email because your Amazon CloudWatch Alarm "IAM-Privilege-Escalation-Alert" in the US East (Ohio) region has entered the ALARM state, because "Threshold Crossed: 1 out of the last 1 datapoints [1.0 (15/05/26 09:00:00)] was greater than or equal to the threshold (1.0) (minimum 1 datapoint for OK -> ALARM transition)." at "Friday 15 May, 2026 09:10:49 UTC".

View this alarm in the AWS Management Console:
<https://us-east-2.console.aws.amazon.com/cloudwatch/deeplink.js?region=us-east-2#alarmsV2:alarm/IAM-Privilege-Escalation-Alert>

Alarm Details:

- Name: IAM-Privilege-Escalation-Alert
- Description:
- State Change: INSUFFICIENT_DATA -> ALARM
- Reason for State Change: Threshold Crossed: 1 out of the last 1 datapoints [1.0 (15/05/26 09:00:00)] was greater than or equal to the threshold (1.0) (minimum 1 datapoint for OK -> ALARM transition).
- Timestamp: Friday 15 May, 2026 09:10:49 UTC
- AWS Account: 922806890303
- Alarm Arn: arn:aws:cloudwatch:us-east-2:922806890303:alarm:IAM-Privilege-Escalation-Alert

Threshold:

- The alarm is in the ALARM state when the metric is GreaterThanOrEqualToThreshold 1.0 for at least 1 of the last 1 period(s) of 300 seconds.

Monitored Metric:

- MetricNamespace: SecurityMonitoring
- MetricName: IAMPrivilegeEscalation
- Dimensions:
- Period: 300 seconds
- Statistic: Sum
- Unit: not specified
- TreatMissingData: missing

State Change Actions:

- OK:
- ALARM: [arn:aws:sns:us-east-2:922806890303:SecurityAlertsbynexwork]
- INSUFFICIENT_DATA:

Step 7 – Splunk Detection

Verify AttachUserPolicy events in Splunk SIEM and create SPL detection queries for privilege escalation monitoring.

New Search Save As Create Table View Close

index=* sourcetype=aws:cloudtrail eventName=AttachUserPolicy Time range: All time +

✓ 13 events (before 5/15/26 2:48:07.000 PM) No Event Sampling Job || → ⬆ ⬇ Smart Mode

Events (13) Patterns Statistics Visualization

Timeline format Zoom Out + Zoom to Selection X Deselect 1 hour per column

Format Show: 20 Per Page View: List

Time	Event
5/15/26 2:41:33.000 PM	<pre>{ "awsRegion": "us-east-1", "eventCategory": "Management", "eventID": "992c4f4b-fd8f-4189-b5d1-edd68a6abfb9", "eventName": "AttachUserPolicy", "eventSource": "iam.amazonaws.com", "eventTime": "2025-05-15T09:11:33Z", "eventType": "AwsApiCall", "eventVersion": "1.1", "managementEvent": true, "readOnly": false, "recipientAccountId": "922868083883", "requestID": "5897a7f9-1668-473b-8873-81fc443b8e46", "requestParameters": { "": {} }, "responseElements": null, "sessionCredentialFromConsole": true, "sourceIPAddress": "47.15.154.224", "tlsDetails": { "": {} } }</pre>

27.010.129000 (cpu:1) [Custom search] [Search]

Splunk Detection Query

index=* sourcetype=aws:cloudtrail eventName=AttachUserPolicy

Conclusion

Successfully implemented AWS Privilege Escalation Detection using AWS CloudTrail, CloudWatch Metric Filters, CloudWatch Alarms, SNS notifications, and Splunk SIEM integration. This workflow enables real-time monitoring and alerting for suspicious IAM policy attachment activities.